M.Kumarasamy College of Engineering
NAAC Accredited Autonomous Institution
Approved by AICTE & Affiliated to Anna University
ISO 9001:2015 Certified Institution
Thalavapalayam, Karur - 639 113, TAMILNADU.

DEPARTMENT OF
ELECTRONICS AND COMMUNICATION ENGINEERING
Academic Year: 2025 - 2026 (Even Sem)

ECB1332 – Design Project
End Semester Review

EVENT-BASED SMART WATER PIPELINE MONITORING AND AUTOMATIC VALVE CONTROL SYSTEM

YEAR/ SEMESTER - III / VI

BATCH NUMBER : 3

DATE : 08.05.2026

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Problem Statement

- Water leakage and theft cause major water loss.
- Traditional systems lack real-time monitoring.
- Manual inspection increases delay and wastage.
- Need for automatic monitoring and valve control.

Objective

- To monitor real-time water flow and pressure in pipelines.
- To detect leakage, theft, and abnormal water usage automatically.
- To classify events intelligently (minor leak, major leak, theft).
- To send alerts to appropriate authorities (municipality / water department).
- To reduce water wastage using automatic valve control.
- To improve overall water management efficiency.

Introduction

- Water is a critical resource, and its efficient management is essential in institutions like colleges.
- Water leakage, theft, and misuse in pipelines lead to major losses.
- Traditional systems lack real-time monitoring and automation.
- Smart technologies like IoT enable continuous monitoring and control.
- This project introduces an event-based smart water pipeline monitoring system.
- It ensures efficient water distribution from main pipelines to sub-pipelines.

Abstract

- Water leakage and unauthorized usage are major issues in campuses.
- Existing systems are mostly manual and inefficient.
- The proposed system uses flow sensors and pressure sensors for monitoring.
- It detects abnormal conditions and classifies events.
- Notifications are sent based on event type.
- Automatic valve control prevents further water loss.
- The system is cost-effective, scalable, and suitable for smart infrastructure.

Existing System

- Manual water meter reading.
- Periodic inspection by maintenance staff.
- No real-time monitoring.
- No detection of theft or misuse.
- No automatic valve control.
- High water wastage.
- Delay in identifying leaks.

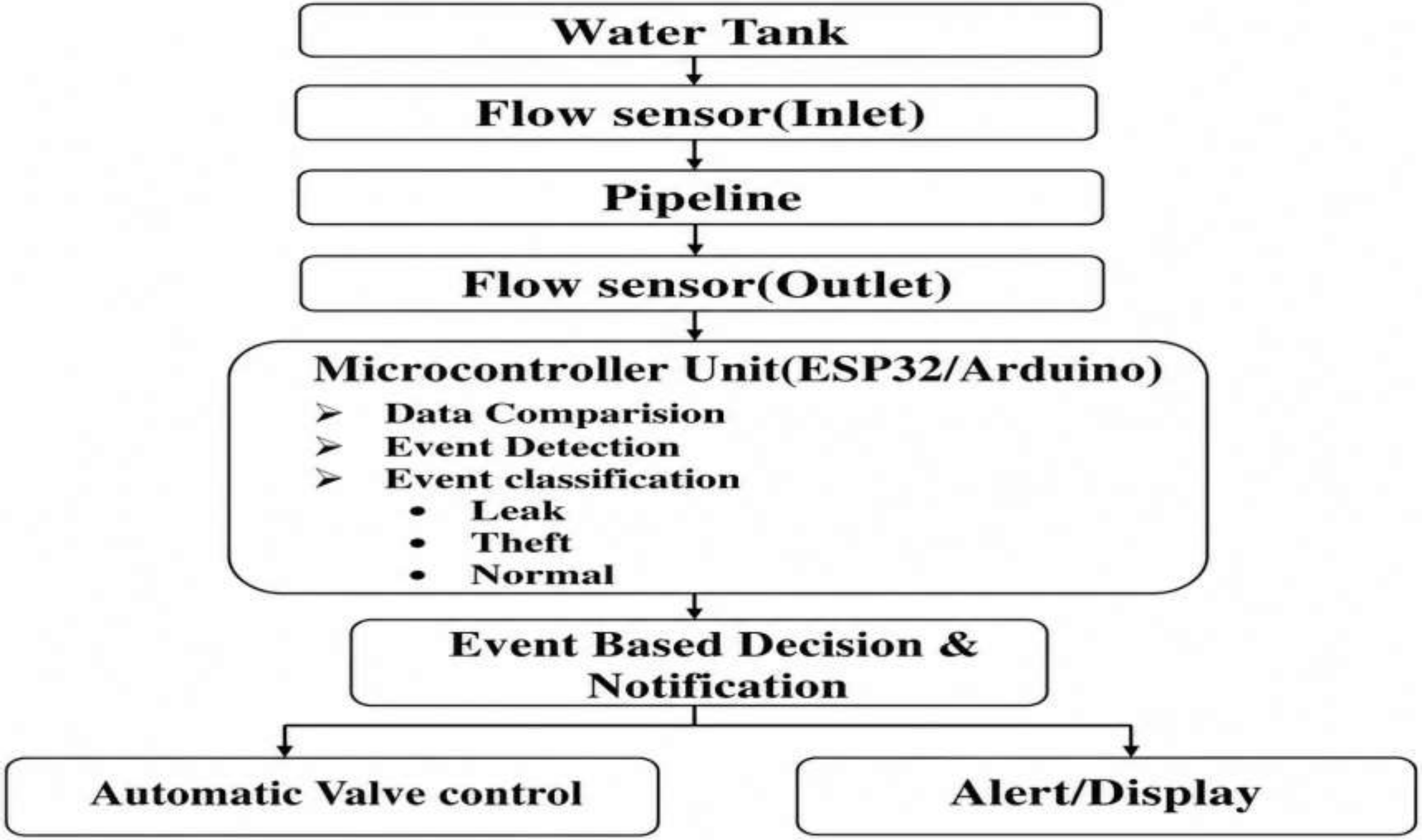
Proposed System

- Sensors continuously monitor pipeline conditions.
- Data is processed using a microcontroller.
- System detects pressure drop and flow variation.
- Events are classified:
  - Minor leak
  - Major leak
  - Theft / abnormal usage
- Solenoid valve closes automatically for major leaks.
- Notifications are sent via IoT.
- Data is stored for analysis.

Literature Review

S.NO	TITLE OF THE PAPER	DESCRIPTION	REMARKS	YEAR
1)	Deep Learning-Based Leak Detection in Water Distribution Systems	Uses deep learning techniques to analyze sensor data and detect leakage patterns in water pipelines	High accuracy but requires large datasets and computational resources	2024
2)	IoT-Based Smart Water Monitoring and Leakage Detection System	Implements IoT sensors for real-time monitoring of water flow and leakage detection	Efficient and scalable; depends on internet connectivity	2023
3)	Acoustic-Based Pipeline Leak Detection Using Signal Processing Techniques	Detects leakage by analyzing acoustic signals generated in pipelines	Suitable for underground pipelines, complex signal processing required	2023
4)	Smart Water Grid: Leak Detection and Localization Using Sensor Networks	Uses distributed sensor networks to detect and locate leaks in water distribution systems	Accurate localization but high installation cost	2023
5)	Real-Time Pipeline Leak Detection Using Pressure and Flow Sensors	Detects leaks by monitoring pressure and flow variations in pipelines	Simple and reliable; less sensitive to very small leaks	2023

Block Diagram



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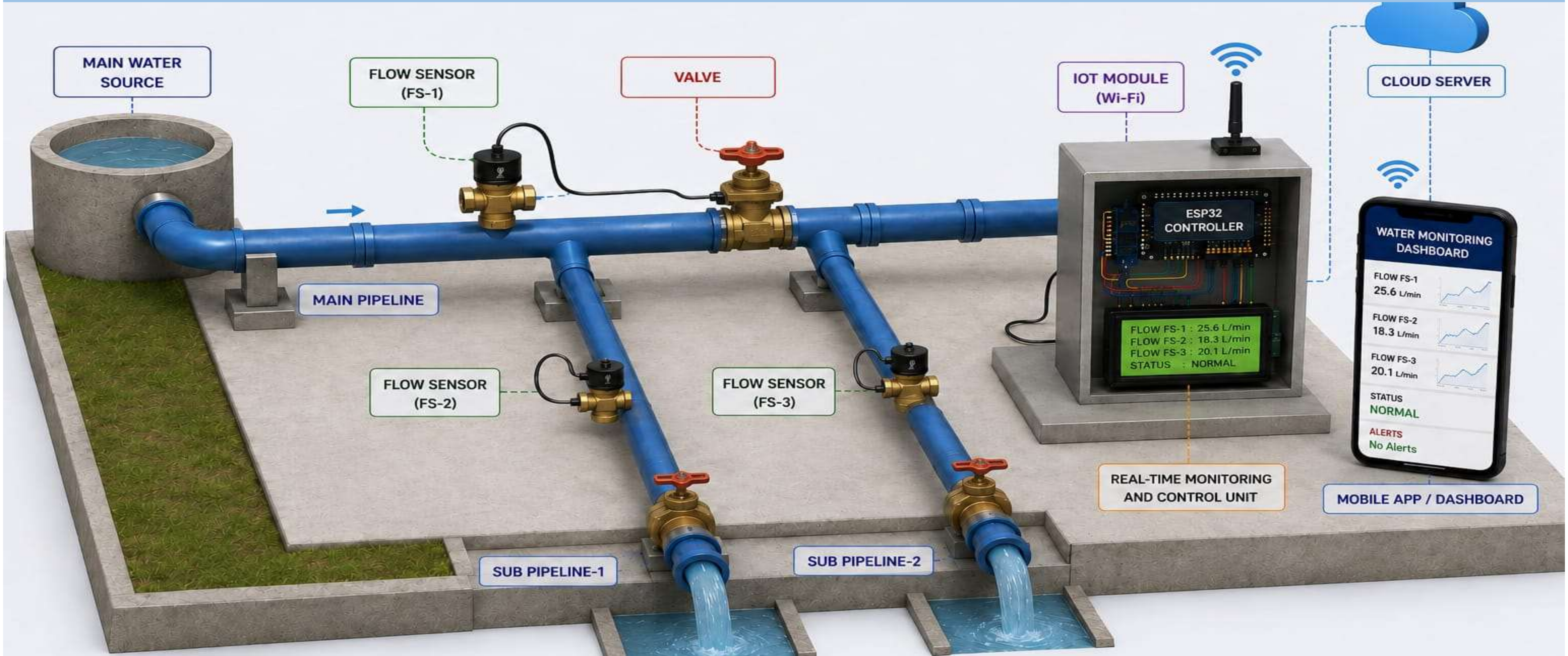
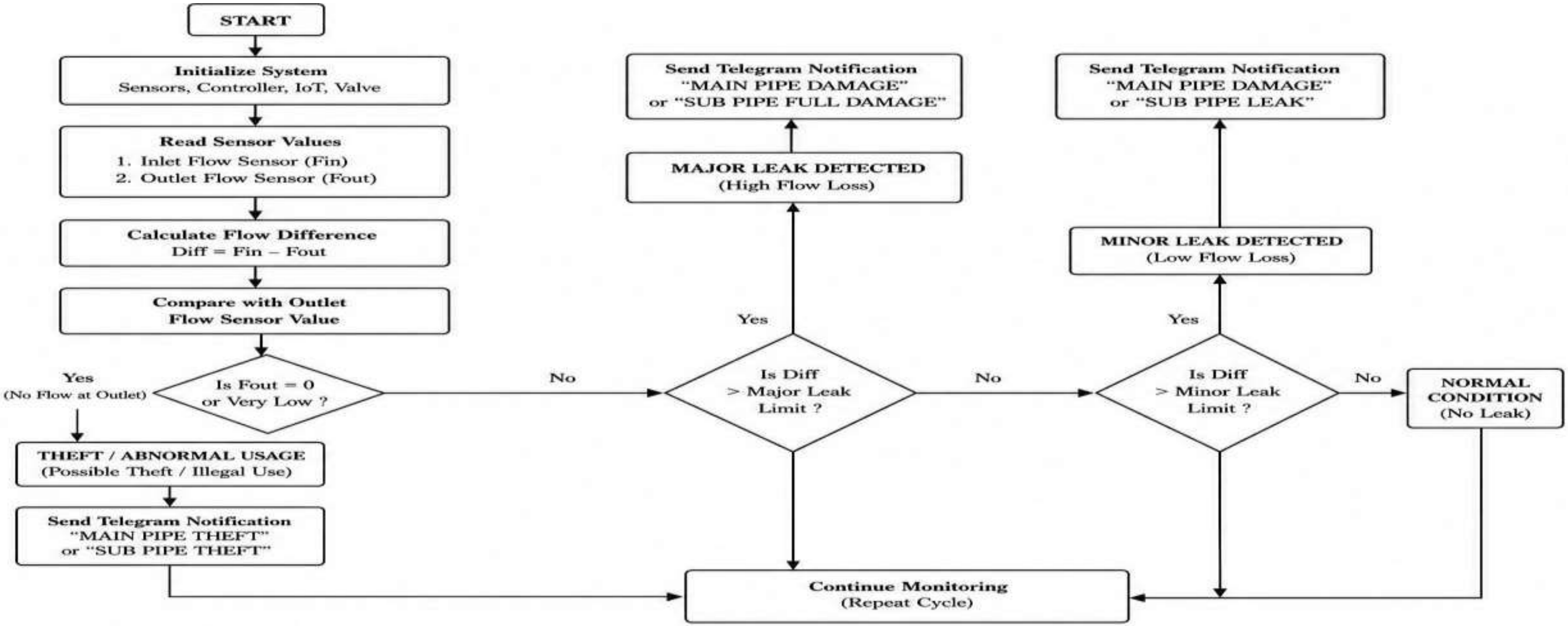
3D Diagram



Flow Chart


START
Initialize System
Sensors, Controller, IoT, Valve
Read Sensor Values
1. Inlet Flow Sensor (Fin)
2. Outlet Flow Sensor (Fout)
Calculate Flow Difference
Diff = Fin – Fout
Compare with Outlet
Flow Sensor Value
Is Fout = 0
or Very Low ?
Yes
(No Flow at Outlet)
THEFT / ABNORMAL USAGE
(Possible Theft / Illegal Use)
Send Telegram Notification
"MAIN PIPE THEFT"
or "SUB PIPE THEFT"
No
Is Diff
> Major Leak
Limit ?
Yes
MAJOR LEAK DETECTED
(High Flow Loss)
Send Telegram Notification
"MAIN PIPE DAMAGE"
or "SUB PIPE FULL DAMAGE"
No
Is Diff
> Minor Leak
Limit ?
Yes
MINOR LEAK DETECTED
(Low Flow Loss)
Send Telegram Notification
"MAIN PIPE DAMAGE"
or "SUB PIPE LEAK"
No
NORMAL
CONDITION
(No Leak)
Continue Monitoring
(Repeat Cycle)

Hardware/Software Used

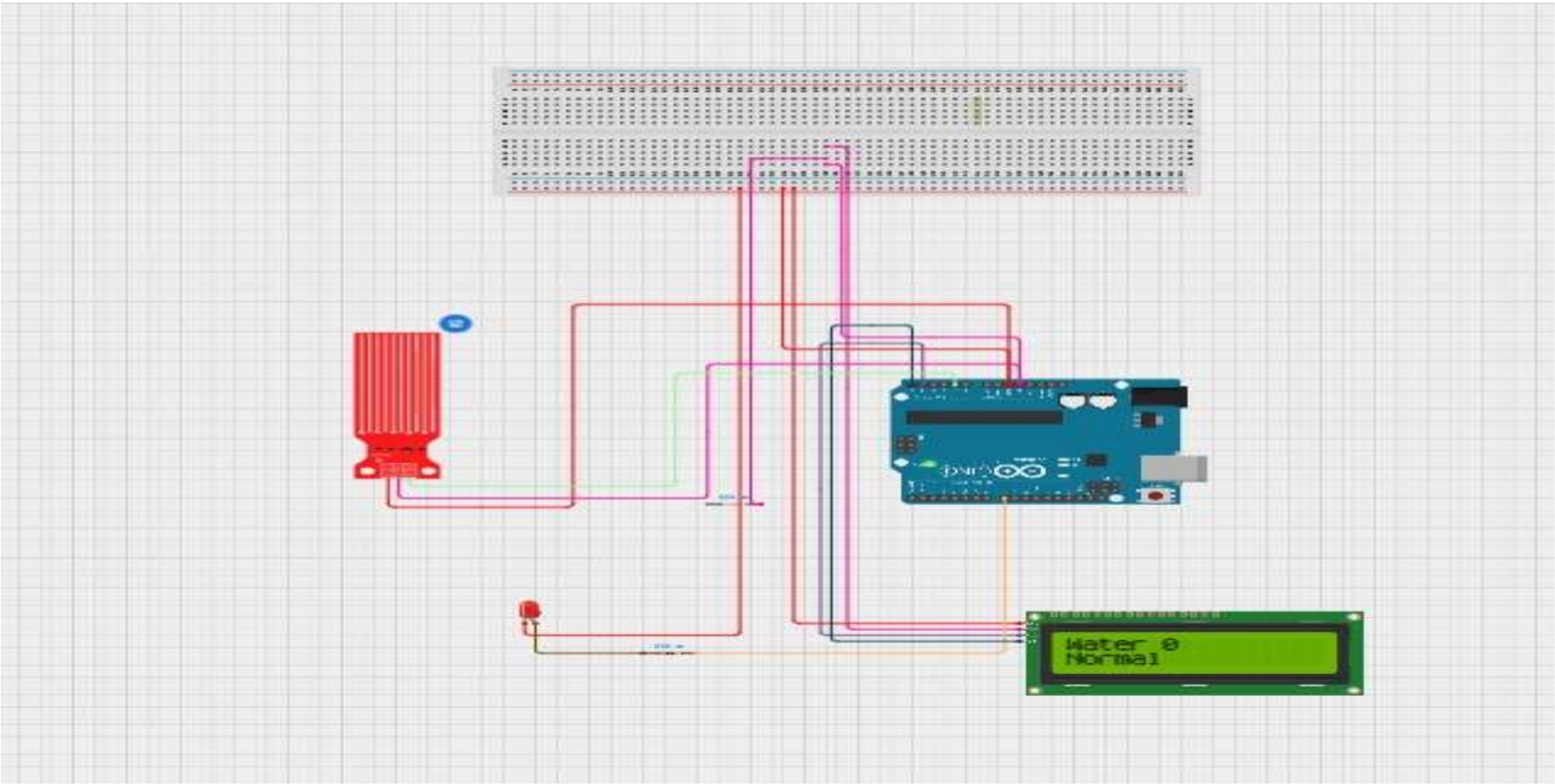
HARDWARE USED:

- ESP32
- Flow Sensor
- Plastic Valve
- PVC Pipe
- Water Storage Tank
- Water Pump
- Adapter
- Ultrasonic Sensor

SOFTWARE USED:

- Telegram
- Webpage

Simulation Results



Hardware Design

Result and Discussion

- System detected leakage successfully.
- Real-time monitoring improved efficiency.
- Automatic valve reduced water wastage.
- IoT alerts enabled quick response.

Applications

- Smart city water systems
- Colleges and campuses
- Residential apartments
- Agricultural irrigation systems

Advantages

- Real-time monitoring
- Automatic leak detection
- Reduces water wastage
- Low maintenance cost
- Fast alert notification

Outcome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Future Scope

- AI-based leak prediction
- Mobile app integration
- Cloud data storage
- Solar-powered system
- Smart city integration

Conclusion

- The system provides an effective solution for water leakage and theft.
- Combines real-time monitoring and intelligent event detection.
- Reduces water wastage and improves efficiency.
- Automatic control minimizes human intervention.
- Scalable and cost-effective solution.
- Suitable for smart infrastructure development.

References

[1] S. Y. Kwon and J. Kim, “A two-phase approach for leak detection and localization in water distribution systems using wavelet decomposition and machine learning,” Computers & Industrial Engineering, vol. 197, 110534, 2024.

[2] Y. Liu, X. Ma, Y. Li, Y. Tie, Y. Zhang and J. Gao, “Water pipeline leakage detection based on machine learning and wireless sensor networks,” Sensors, vol. 19, no. 23, 2019.

[3] P. Irofti, L. Romero-Ben, F. Stoican and V. Puig, “Factor graph optimization for leak localization in water distribution networks,” arXiv:2509.10982, 2025.

[4] O. Melnikov, Y. Dorofieiev, Y. Shakhnovskiy, H. Truong and V. Degeler, “A multivariate statistical framework for detection, classification and pre-localization of anomalies in water distribution networks,” arXiv:2512.15685, 2025.

Thank You!!!